

Time to normalization of gait following ACL reconstruction compared with healthy controls: A systematic review and meta-analysis

Shiwei Chen^{a,1} , Han Gong^{b,2}, Chennan Lyu^{c,3}, Kehan Li^{d,4}, Shazlin Shaharudin^{a,*,5}

^a School of Health Sciences, Universiti Sains Malaysia, Kelantan, Malaysia

^b School of Biological Science and Medical Engineering, Beihang University, Beijing Key Lab for Biomaterials and Neural Regeneration, Beijing, China

^c School of Physical Education, Wuhan Sports University, Wuhan, China

^d Department of Sports Studies, Universiti Putra Malaysia, Selangor, Malaysia

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ABSTRACT

Background: Gait abnormalities often persist after anterior cruciate ligament reconstruction (ACLR) and may increase the risk of reinjury and joint degeneration. This review aimed to assess differences in key gait parameters between ACLR patients and healthy controls, and to estimate when these parameters normalize postoperatively.

Methods: Studies comparing individuals ≥ 3 months post-ACLR with healthy controls were included. A systematic search across five databases yielded 5251 records, with 20 studies ($n = 976$) retained. A longitudinal multilevel meta-analysis was performed, with pooled effect sizes calculated using Cohen's d , representing the standardized mean difference. Postoperative time was modeled as a moderator in meta-regression analyses.

Results: Compared to healthy controls, ACLR patients assessed at 3–107 months postoperatively exhibited significantly lower peak knee flexion angle ($d = -0.48$, 95 % CI: -0.87 to -0.10) and peak knee flexion moment ($d = -1.06$, 95 % CI: -2.06 to -0.07), while walking speed was non-significant ($d = -0.17$, 95 % CI: -0.47 – 0.13). Logarithmic modeling indicated that group differences became non-significant at 16.2 months for flexion angle and 10.1 months for flexion moment. Walking speed was statistically non-significant at any timepoint.

Significance: At ≥ 3 months post-ACLR, patients exhibited significantly reduced peak knee flexion angle and flexion moment compared to controls. These deficits normalized at approximately 16.2 and 10.1 months, respectively. Walking speed did not significantly differ at any postoperative timepoint. These findings suggest that early-phase rehabilitation should emphasize quadriceps strengthening and knee flexion restoration within the first 10–16 months to address persistent biomechanical deficits.

1. Introduction

Injuries to the anterior cruciate ligament (ACL) represent one of the most frequent and serious forms of knee trauma in athletic populations [1]. Epidemiological data indicate that tens of thousands of ACL ruptures occur annually, with an estimated incidence of 1 in 3500 per year in the United States alone [2]. ACL injuries often lead to joint pain,

instability, and early-onset osteoarthritis, significantly impairing physical performance and quality of life [3].

Anterior cruciate ligament reconstruction (ACLR) is currently the treatment of choice for restoring knee stability, with the primary objective of enabling patients—particularly young athletes—to safely return to sport [4]. Although most patients achieve clinical stability following ACLR, gait recovery often remains incomplete in the short

* Correspondence to: School of Health Sciences, Universiti Sains Malaysia, Kubang Kerian, Kelantan 16150, Malaysia.

E-mail addresses: shiweichenc@student.usm.my (S. Chen), gonghan2024@buaa.edu.cn (H. Gong), 2023420028@whsu.edu.cn (C. Lyu), gs70726@student.upm.edu.my (K. Li), shazlin@usm.my (S. Shaharudin).

¹ ORCID: 0009-0005-2919-4667.

² ORCID: 0009-0006-1466-7430.

³ ORCID: 0009-0008-8675-2057.

⁴ ORCID: 0009-0002-5085-4189.

⁵ ORCID: 0000-0001-5583-7631.

term [5]. Research has demonstrated that ACLR patients commonly exhibit biomechanical impairments during gait compared to healthy controls [6], with persistent gait abnormalities frequently observed even at 6–12 months postoperatively, and in some cases lasting up to two years [7]. Such gait abnormalities not only increase the risk of ACL re-injury but are also closely associated with postoperative joint degeneration [7,8]. Therefore, identifying which gait parameters are prone to long-term impairment and which parameters recover relatively early following surgery is critical for evaluating rehabilitation progress and developing targeted intervention strategies.

Currently, the findings regarding gait recovery following ACLR remain inconsistent. For example, one study reported that ACLR patients exhibited lower external knee flexion moment during walking compared to healthy controls between 2 and 12 months postoperatively [9], while another found that both knee flexion angle and joint moments were higher in ACLR patients within the first 6 months post-surgery [10]. Although numerous studies have focused on alterations in joint loading patterns during gait after ACLR, systematic summaries specifically addressing the recovery timeline of gait parameters postoperatively remain limited [11].

Therefore, a comprehensive synthesis of existing literature is necessary to determine the critical time points at which key gait parameters return to normal following ACLR, which is critical for optimizing rehabilitation and guiding clinical assessment timing. Meta-analysis provides a robust statistical framework to synthesize findings across studies and to identify time-specific recovery trajectories following ACL reconstruction [12], thereby making it well-suited to address this question.

However, most existing meta-analyses have utilized traditional frameworks, treating each study as an independent unit and extracting effect sizes only at single time points [13], thus failing to fully leverage data available from multiple follow-up assessments [14,15]. Consequently, such approaches are insufficient to characterize the dynamic trajectory of postoperative gait recovery.

To address these limitations, the present study aims to employ a multilevel longitudinal meta-analytic approach. This method moves beyond the traditional meta-analytic assumption of single effect-size extraction by integrating data across multiple follow-up time points, thus providing detailed insight into the temporal patterns of gait recovery. Given that current research on postoperative gait after ACLR primarily focuses on core parameters such as walking speed, peak knee flexion angle, and peak knee flexion moment [16–18], and considering that consensus regarding their recovery trajectories has not yet been reached [19], this study will specifically examine the dynamic changes of these three parameters. Clarifying the time points at which these parameters return to normal baseline value will provide robust evidence to inform the optimization of postoperative rehabilitation strategies and the establishment of clinical evaluation frameworks.

2. Methods

This systematic review and meta-analysis were conducted following the PRISMA guidelines [20]. Additionally, the study protocol was registered with the PROSPERO database (registration number: CRD420251073937) on June 14, 2025.

2.1. Eligibility criteria

Participants included in this study were patients with primary ACL rupture who underwent surgical reconstruction and had a postoperative recovery period of at least three months. Each study needed to include a healthy control group, ensuring baseline comparability between ACLR patients and healthy individuals; comparisons must be made between ACLR patients and healthy controls (i.e., between-subject comparisons). Studies comparing reconstructed ACL limbs with the contralateral healthy limbs (within-subject comparisons) were excluded to avoid bias

due to individual variability.

Studies had to report quantitative data for at least one of the following gait-related parameters: peak knee flexion angle, peak knee flexion moment, or walking speed. Additionally, studies needed to explicitly identify the reconstructed limb.

2.2. Literature search and selection

After reviewing key literature to test appropriate terminology, our search specifically targeted two elements: population (anterior cruciate ligament reconstruction or ACL reconstruction or anterior cruciate ligament or ACL) and outcomes ("gait" or "walking"). We systematically searched PubMed, Embase, The Cochrane Library, Web of Science, and EBSCOhost-SportDiscus using appropriate Medical Subject Headings (Mesh) terms tailored to each database from database inception up to February 28, 2025 (see Supplementary File 1). Additionally, relevant meta-analyses on this topic were cross-referenced to identify further original studies meeting our inclusion criteria.

2.3. Data extraction

Two independent pairs of reviewers (S.C. & H.G.; C. Lyu. & K.L.) extracted data according to a customized data extraction table. Information extracted included study design, basic demographic characteristics (e.g., sample size, country, age, percentage of females, body mass index, patient population), postoperative time points, and gait-related outcome data (means and standard deviations for both experimental and control groups). Since this was a longitudinal meta-analysis comparing ACLR patients with healthy controls, all available time points from long-term studies were extracted.

Upon completion of data extraction, the two reviewer pairs (S.C. & H.G.; C. Lyu. & K.L.) performed cross-validation together with the corresponding author (S.S.). Any inconsistencies were resolved through discussion within the research team, and relevant data were re-extracted as necessary. Finally, the corresponding author (S.S.) conducted a comprehensive review of all extracted data to ensure accuracy and consistency.

2.4. Risk of bias and grade assessment

All studies included in this research were observational, comparing gait parameters of ACLR patients with healthy controls (non-randomized controlled trials studies). Two pairs of reviewers (S.C. & H.G.; C. Lyu. & K.L.) independently assessed the risk of bias using the Risk of Bias In Non-randomized Studies of Interventions (ROBINS-I) tool, evaluating seven bias domains [21]. Any disagreements were resolved through discussion with the corresponding author (S.S.).

To evaluate the overall certainty of evidence, we conducted a structured evidence grading process using the GRADE (Grading of Recommendations Assessment, Development and Evaluation) framework, in accordance with the GRADE Working Group's guidance for non-randomized studies [22]. We systematically assessed five domains—risk of bias, inconsistency, indirectness, imprecision, and publication bias—for each outcome. Detailed outcome-specific GRADE ratings are presented in [Supplementary File 2](#), indicating low to very low certainty of evidence for peak knee flexion angle, peak knee flexion moment, and walking speed.

2.5. Publication bias and sensitivity analysis

Funnel plot asymmetry and Egger's regression test were used to evaluate publication bias, with $p > 0.05$ indicating no evidence of significant bias [23]. Given that the longitudinal multilevel meta-analysis incorporated multiple follow-up time points within individual studies, we evaluated publication bias at two levels: first, within individual studies (assessing data symmetry); second, across studies (evaluating

overall distribution symmetry) [24]. Sensitivity analyses were conducted by systematically excluding each study to assess its individual influence on the overall results, analyzing both within and between studies.

2.6. Statistical analysis

To evaluate the temporal trends in gait parameters following ACL reconstruction, we conducted a series of meta-analyses using RStudio. First, pooled standard deviations (pooled SD) were calculated from the means and standard deviations reported by experimental and control groups. From these, standardized mean differences (SMD, Cohen's *d*) and their variances were computed to quantify effect sizes and statistical uncertainty [25]. To account for correlation between multiple time points within the same study, we assumed a correlation coefficient of $r = 0.85$, consistent with previous longitudinal meta-analyses in ACLR patients [26]. This value was used to construct a within-study variance-covariance matrix for the multilevel meta-analysis model. Cohen's *d* values were classified according to Cochrane Handbook thresholds: < 0.2 as trivial, $0.2-0.5$ as small, $> 0.5-0.8$ as moderate, and > 0.8 as large [27], facilitating comparisons between ACLR patients and healthy controls at different time points.

A longitudinal mixed-effects meta-analysis was performed using the restricted maximum likelihood (REML) method, primarily employing the metafor package in R, following established analytical frameworks [28,29]. To handle potential correlations across multiple follow-up time points within different study cohorts, we developed multilevel correlated-effect models [30]. The final model defined time points as random effects nested within individual study cohorts and incorporated a continuous autoregressive (CAR) structure to model temporal correlations.

Subsequently, time points were included as fixed-effect moderators to analyze trends in gait parameters over time. To identify the best model fit, we explored various modeling strategies, including linear,

logarithmic-linear, 3-knot spline, and 4-knot spline models. Optimal model selection was guided by criteria including Akaike Information Criterion (AIC), Bayesian Information Criterion (BIC), and visual inspection of fitted curves. Spline modeling was performed using the rms package, with knot placement based on recommended guidelines [31]. Robust variance estimation for the final model was conducted using the clubSandwich package [22].

The final results were presented as predicted effect sizes with their 95 % confidence intervals, quantifying both magnitude and statistical uncertainty. Additionally, the predict() function was used to estimate effects at critical time points and to determine the time points at which the effect sizes reached zero (i.e., statistical significance between ACLR and control groups disappeared), thereby identifying critical milestones for gait parameter recovery [32].

Detailed information about model specifications, parameters, scripts, and all data is publicly available at: <https://github.com/Shiwei-Lab/ACLR-Gait-Meta2025>.

3. Results

3.1. Study selection and basic characteristics

As of February 28, 2025, a total of 5251 records were retrieved through database searches. After initial screening, 70 potentially eligible studies were identified. Following full-text review, 20 studies [11,14,15,17,33–47,59] were ultimately included in the meta-analysis. A detailed flowchart outlining the study selection process is presented in Fig. 1.

These 20 studies collectively involved 976 participants. All included studies compared gait parameters between individuals with ACLR and healthy controls, focusing primarily on peak knee flexion angle, peak knee flexion moment, and walking speed. Postoperative follow-up periods for gait assessments ranged from 3 m to 107 months. Notably, 8 studies reported gait outcomes at multiple postoperative time points. Key characteristics of the included studies are summarized in Table 2.

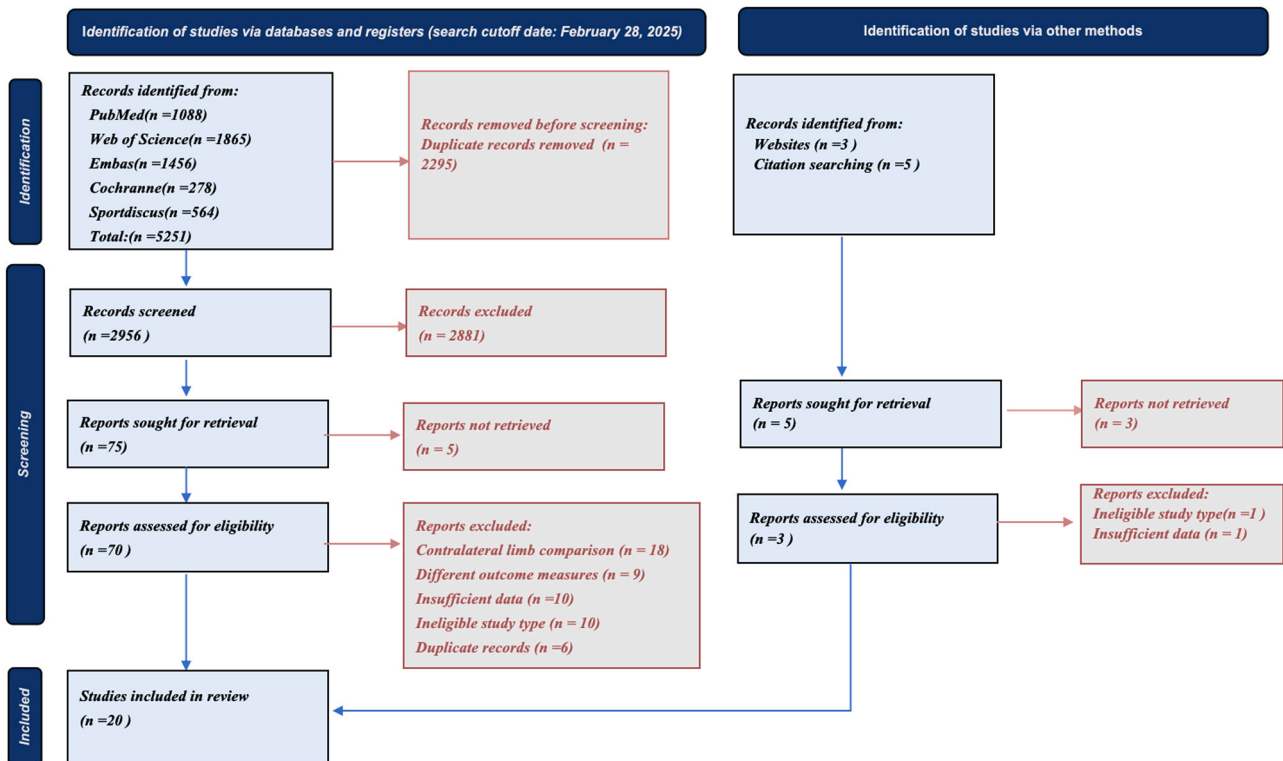


Fig. 1. PRISMA flow diagram of study selection. Flowchart outlining the literature screening process, including the number of records identified, screened, excluded (with reasons), and the final studies included in the meta-analysis.

Table 1

Estimated gait parameter differences at 3, 6, and 12 months after ACL reconstruction.

Gait test	N (total sample)	Total studies [total effects (k)]	3-month estimate	6-month estimate	12-month estimate	Estimated time- point of no difference (months)
Peak Knee Flexion Angle	401	9(11)	−0.75(−1.33, −0.18) *	−0.61(−1.04, −0.18) *	−0.46 (−0.84, −0.08)	16.2
Peak Knee Flexion Moment	230	5(7)	−1.87(−3.18, −0.567) *	−1.29(−2.17, −0.42) *	−0.71(−1.63, 0.21)	10.1
Walking Speed	486	10(13)	−0.28(−0.75, 0.19)	−0.23(−0.60, 0.14)	−0.19(−0.50, 0.13)	N/A

Estimated time-point of no difference (months): The predicted postoperative time at which the 95 % confidence interval for the effect estimate first crosses zero (i. e., no difference between groups). N/A: Not Applicable; indicates that the confidence interval did not cross zero within the observed time range.

Table 2

Characteristics of included literature.

Author	Country	N (total sample)	Measure	Post-operative time	Age(years)	Height(m)/	Weight(kg)/ BMI(Kg/m)
Georgoulis,2003	Greece	31	gait velocity (m/s) Peak Knee flexion angle (°)	7 months	C:24.7 ± 3.7 E:25 ± 4	C:1.73 ± 0.05 E:1.73 ± 0.06	C:62.1 ± 12.4 kg E:69.1 ± 7.9 kg
Czamara,2015	Poland	62	Peak Knee flexion angle (°)	3.2 months	C:25.93 ± 4.57 E: 28.65 ± 8.57	C:1.81 ± 0.07 E:1.81 ± 0.09	C:80.4 ± 12.3 E:81.9 ± 11.7
Padhye,2023	USA	60	Averaged gait velocity (m/s)	48 months	C:22.7 ± 2.9 E:22.3 ± 4.4	C:73 ± 15.7 E:73.1 ± 15.7	N/A
Noehren,2013	USA	40	Peak Knee flexion angle (°)	62 months	C:26 ± 5.1 E:25 ± 6.2	C:1.6 ± 0.06 E:1.6 ± 0.06	C:61 ± 5.1 E:64 ± 8.2
Büttner,2024	USA	116	Gait speed (m/s)	3.76, 5.74, 11.4 months	C:21.6 ± 4.3 E: 21.8 ± 4.7	N/A	C: BMI:24.5 ± 3.2 E: BMI:24.2 ± 3.2
Armitano-Lago,2020	USA	32	Velocity (m/s)	107 months	C:28.90 ± 6.24 E:29.25 ± 6.9	C:1.71 ± 0.09 E: 17.4 ± 0.07	C:74.61 ± 17 E:81.46 ± 14
Skvortsov,2020	Russia	40	Peak Knee flexion angle (°)	12 months	C:34.2 E:29.7	N/A	N/A
Leporace,2016	Brazil	22	Gait Velocity (m/s)	11 months	C:27.3 ± 2.7 E:33.1 ± 11.1	C:1.80 ± 0.04 E: 1.82 ± 0.03	C:82.1 ± 9.5 E:82.1 ± 7.4
Davis-Wilson,2019	USA	60	Walking speed (m/s)	6, 12 months	C:22 ± 4 E:22 ± 3	N/A	C: BMI:23.6 ± 2.1 E: BMI:23.8 ± 2.2
Goetschius,2018	USA	58	Peak Knee flexion motion (°)	103 months	C:22.4 ± 3.2 E:26.7 ± 4.4	C:1.71 ± 0.13 E: 1.73 ± 0.1	C:68.9 ± 13.1 E:69.5 ± 12.7
Davis,2018	USA	40	Walking Speed (m/s)	84 months	N/A	N/A	C: BMI:22.4 ± 2.4 E: BMI: 25 ± 3.5
Hadizadeha,2016	Malaysia	37	Mean gait velocity (m/s)	3 months	C:21.5 ± 1.0 E:23.6 ± 5.4	C:1.66 ± 0.07 E:1.69 ± 0.09	C:61.7 ± 11.3 E:66 ± 13.8
Patterson,2013	Ireland	34	Gait velocity (m/s)	42 months	C:20.8 ± 1.17 E:23.7 ± 3.12	C: 1.65 ± 0.06 E: 1.64 ± 0.05	C:64.7 ± 7.06 E:64.9 ± 9.02
Zabala,2013	USA	90	Peak Knee Flexion Momen (Nm/kg)	26 months	C:30.2 ± 4.68 E:29.5 ± 6.1	C: 1.73 ± 0.09 E: 1.72 ± 0.10	C:74.2 ± 12.3 E:74.4 ± 12.4
Lewek,2002	USA	28	Peak Knee Flexion moment Peak Knee flexion angle (°) Peak Knee Flexion (°)	3.3, 4.8 months	C:32.2 ± 6.6 E: 25 ± 7.8	N/A	N/A
Michelle Hall,2012	Australia	32	Peak Knee flexion angle (°)	72 months	C:26 ± 4 E: 26 ± 6	C: 1.7 ± 0.12 E: 1.74 ± 0.14	C:68 ± 2 E:75 ± 15
Karimi,2013	Iran	30	Gait speed (m/s) Peak Knee Flexion moment (Nm/kg)	6 months	C:32 ± 3 E: 33 ± 2.6	C:1.74 ± 0.06 E:1.75 ± 0.05	N/A
Reed Ferber,2002	USA	20	Peak Knee flexion moments (Nm/kg)	3, 4 months	C:24.4 ± 3.1 E:27.7 ± 9.1	C: 1.7 ± 0.09 E: 1.66 ± 0.2	C:67.2 ± 10.7 E:79.1 ± 13.8
Kono,2023	Japan	62	Peak Knee flexion angle (°) Peak knee flexion moment (Nm/kg/m)	12 months	C:24.07 ± 4.73 E:25.91 ± 11.61	C: 1.68 ± 0.09 E: 1.66 ± 0.09	C:63.5 ± 12.4 E:64.1 ± 12.6
Zsolt Knoll,2004	Hungary	76	Peak Knee flexion angle (°)	8, 12 months	C:31.70 ± 4.1 E:34.78 ± 6.91	C: 1.71 ± 0.12 E: 1.73 ± 0.57	C:72.1 ± 25.2 E:84.7 ± 15.6

Abbreviations: C, control group; E, experimental group.

3.2. Risk of bias and publication bias

An overview of the risk of bias across included studies is provided in Supplementary File 3. The majority of studies demonstrated methodological robustness in terms of intervention classification, handling of

missing data, outcome measurement, and selective reporting. However, the quality of evidence was primarily affected by confounding factors and selection bias. Specifically, only one study was rated as having a low risk of bias [33]. Sixteen studies were judged to have a moderate risk of bias, largely due to acceptable limitations in participant selection. Three

studies were assessed as high risk [41,45,59], primarily due to inadequate control of confounders and participant selection issues.

Regarding publication bias, funnel plots and Egger's regression tests are shown in [Supplementary File 4](#). Under the conventional two-level modeling framework, both peak knee flexion moment and walking speed outcomes demonstrated marked asymmetry, with significant Egger test results ($p < 0.001$), indicating left-skewed and right-skewed distributions, respectively. In contrast, the distribution for peak knee flexion angle appeared more symmetrical, with a non-significant Egger test ($p = 0.31$), suggesting no substantial publication bias.

Under the multilevel meta-analytic framework, distributions for peak knee flexion angle and peak knee flexion moment remained symmetrical, with non-significant Egger test results ($p = 0.54$ and $p = 0.07$, respectively), indicating no notable publication bias. For walking speed, however, Egger's test yielded a significant result ($p = 0.04$), suggesting potential but not definitive publication bias. Thus, evidence for publication bias in walking speed outcomes remains limited.

3.3. Main effects of gait parameters

3.3.1. Peak knee flexion angle

ACLR patients exhibited significantly lower peak knee flexion angles during walking compared to controls (SMD = -0.48 , 95 % CI: -0.87 to -0.10^* , $p = 0.013$). Heterogeneity analysis revealed substantial between-study heterogeneity ($\tau^2 = 0.23$, $\rho = 0.90$, $Q = 25.55$, $p = 0.004$). Detailed results are presented in [Supplementary File 5A](#) (forest plot and related model outputs).

3.3.2. Peak knee flexion moment

For peak knee flexion moment, ACLR patients showed significantly lower values compared to healthy controls (SMD = -1.06 , 95 % CI: -2.06 to -0.07^* , $p = 0.04$). Heterogeneity analysis again indicated considerable between-study variability ($\tau^2 = 1.15$, $\rho = 0.88$, $Q = 29.20$, $p < 0.001$), as shown in [Supplementary File 5B](#).

3.3.3. Walking speed

For walking speed, ACLR patients exhibited slightly reduced values compared to controls, but the difference did not reach statistical significance (SMD = -0.17 , 95 % CI: -0.47 – 0.13 , $p = 0.28$). Heterogeneity was significant across studies ($\tau^2 = 0.14$, $\rho = 1.00$, $Q = 27.46$, $p = 0.01$), as shown in [Supplementary File 5C](#).

3.4. Meta-regression results with postoperative time as a moderator

This analysis examined the relationship between postoperative time and gait recovery by treating time as a moderator variable. The best-fitting models were then used for final estimation.

3.4.1. Peak knee flexion angle

The effect of postoperative time on peak knee flexion angle was assessed using a logarithmic model. Results indicated that time since surgery was not a statistically significant moderator of peak knee flexion angle ($F = 1.99$, $p = 0.24$). Although the trend suggested gradual improvement in peak knee flexion angle over time, the regression coefficient did not reach statistical significance ($k = 11$, $\beta = 0.21$, 95 % CI: -0.23 – 0.65 , $p = 0.24$). As shown in [Fig. 2\(A\)](#).

3.4.2. Peak knee flexion moment

A logarithmic model was also employed to assess the influence of postoperative time on peak knee flexion moment. The results similarly showed no statistically significant moderating effect of time ($F = 2.02$, $p = 0.28$). While the overall trend indicated some improvement in peak knee flexion moment over time, the association was not statistically significant ($k = 7$, $\beta = 0.84$, 95 % CI: -1.44 – 3.12 , $p = 0.28$). As shown in [Fig. 2\(B\)](#).

3.4.3. Walking speed

The relationship between postoperative time and walking speed was evaluated using a linear model. The analysis revealed no statistically significant trend ($F = 0.14$, $p = 0.73$). Although walking speed appeared to improve slightly over time, the regression coefficient remained non-significant ($k = 13$, $\beta = 0.002$, 95 % CI: -0.018 – 0.023 , $p = 0.73$). As shown in [Fig. 2\(C\)](#).

3.5. Trajectory of gait recovery at key postoperative time points

Gait recovery trajectories were further examined at five key postoperative time points (3, 6, 12, 18, and 24 months) using model-based predictions. Although spline and logarithmic models yielded similar AIC/BIC values, the logarithmic model was ultimately selected for its greater interpretability. The log function captures the nonlinear trend of early rapid recovery followed by a plateau, which aligns with known rehabilitation trajectories in ACLR patients. Additionally, it reduces the risk of overfitting relative to spline models given the moderate number of studies available at later time points.

3.5.1. Peak knee flexion angle

At 3 months postoperatively, ACLR patients exhibited significantly lower peak knee flexion angles compared to healthy controls (SMD = -0.75 , 95 % CI: -1.33 to -0.18). This difference gradually diminished over time: at 6 months (SMD = -0.61 , 95 % CI: -1.04 to -0.18); 12 months (SMD = -0.46 , 95 % CI: -0.84 to -0.08); 18 months (SMD = -0.38 , 95 % CI: -0.79 – 0.03); and 24 months (SMD = -0.32 , 95 % CI: -0.78 – 0.14). According to model-based predictions, the 95 % confidence interval for the effect estimate first crossed zero at approximately

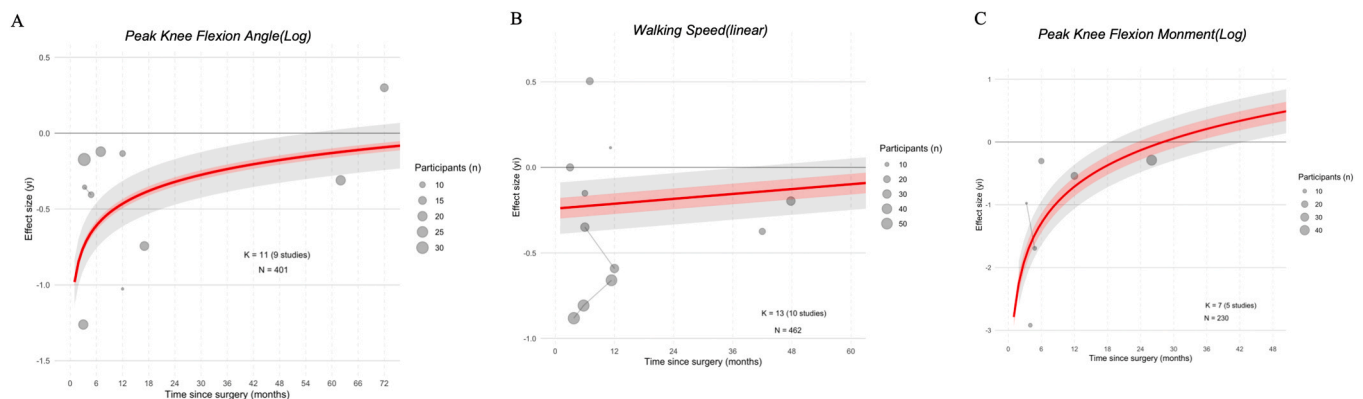


Fig. 2. Meta-regression of gait over time. Meta-regression plot showing the relationship between postoperative time and standardized mean differences (SMD) in gait parameters. The curve represents the fitted longitudinal mixed-effects model with restricted maximum likelihood (REML) estimation. Shaded areas indicate 95 % confidence intervals.

16.2 months, indicating that group differences in peak knee flexion angle were no longer statistically significant beyond this time point. Detailed results are provided in Table 1 and Fig. 3(A).

3.5.2. Peak knee flexion moment

At 3 months postoperatively, peak knee flexion moments in ACLR patients were significantly lower than in healthy controls (SMD = -1.87 , 95 % CI: -3.18 to -0.56). This difference lessened over time: at 6 months (SMD = -1.29 , 95 % CI: -2.17 to -0.42); 12 months (SMD = -0.71 , 95 % CI: -1.63 – 0.21); 18 months (SMD = -0.37 , 95 % CI: -1.54 – 0.79); and 24 months (SMD = -0.13 , 95 % CI: -1.52 – 1.26). According to the model's fitted predictions, the confidence interval for the effect estimate crossed zero at approximately 10.1 months postoperatively, suggesting that group differences in peak knee flexion moment were no longer statistically significant after this time point. Full results are shown in Table 1 and Fig. 3(B).

3.5.3. Walking speed

At 3 months postoperatively, ACLR patients demonstrated slightly reduced walking speed relative to healthy controls (SMD = -0.28 , 95 % CI: -0.75 – 0.19). This trend of minimal difference persisted over time: at 6 months (SMD = -0.23 , 95 % CI: -0.60 – 0.14); 12 months (SMD = -0.19 , 95 % CI: -0.50 – 0.13); 18 months (SMD = -0.16 , 95 % CI: -0.47 – 0.15); and 24 months (SMD = -0.14 , 95 % CI: -0.46 – 0.18). At all key time points, the 95 % confidence intervals crossed zero, indicating that while walking speed showed a trend toward improvement over time, none of the observed differences reached statistical significance. Detailed data are presented in Table 1 and Fig. 3(C).

4. Discussion

4.1. Analysis of differences in gait parameters

4.1.1. Peak knee flexion angle differences

The results of this review indicate that ACLR patients exhibited significantly reduced peak knee flexion angles during walking compared to healthy controls, consistent with findings from existing systematic reviews. Previous reviews reported generally lower peak knee flexion angles in ACL-reconstructed patients compared to healthy individuals during the postoperative period [53], and similar deficits have also been reported during running at various stages of postoperative recovery [62, 63, 65]. This limitation in flexion angles may be attributed to excessive scarring and adhesions on the medial and lateral sides of the patella following surgery [54]. Additionally, psychological factors such as patients' fear of reinjury may reduce patients' willingness or ability to fully flex the knee [55]. Similarly, recent findings suggest that quadriceps strength is associated with differences in knee flexion angle during gait among individuals with ACLR [61], suggesting a possible link between muscle function and knee flexion angle. Collectively, these factors likely

restrict the range of motion, thereby affecting knee flexion performance during gait.

4.1.2. Peak knee flexion moment differences

Our review found that peak knee flexion moments were significantly lower in ACLR patients compared to healthy controls, aligning with trends reported in prior literature. Previous reviews highlighted that reduced peak knee flexion moments commonly occur among ACL-reconstructed patients for a period post-surgery [56]. This reduction in peak knee flexion moment is potentially due to early postoperative muscle atrophy, impaired neuromuscular control, and protective compensatory strategies to avoid loading the reconstructed limb [57, 58]. These combined factors likely contribute to the observed deficit in peak knee flexion moment during gait.

4.1.3. Walking speed differences

The present review noted that ACLR patients exhibited slightly lower walking speeds compared to healthy controls, although the difference was not statistically significant. This observation aligns with certain previous studies, which reported no significant differences in spatio-temporal gait parameters, including walking speed, between ACLR patients and healthy controls within 4–15 months post-surgery [59]. This phenomenon might be explained by patients' subjective adjustments and compensatory mechanisms involving other joints [60], allowing them to maintain nearly normal walking speeds despite incomplete knee function recovery.

4.2. Recovery trends of gait parameters over time

4.2.1. Temporal changes in peak knee flexion angle

At 3 months postoperatively, ACLR patients showed significantly lower knee flexion angles compared to healthy controls, indicating noticeable asymmetry in sagittal plane movement between affected and unaffected limbs in early postoperative gait [48]. Over time, these differences gradually diminished and became statistically non-significant at approximately 16.2 months, suggesting recovery of peak knee flexion angle by the mid-term postoperative stage. This recovery trend is consistent with previous research; for example, studies have shown that both males and females exhibit abnormal peak knee flexion angles at 6 months post-ACLR, with normalization typically observed around 12 months [49]. However, some cross-sectional evidence suggests that even in the later stages of recovery—on average four years after surgery—ACLR patients may still exhibit reduced knee flexion angles during running compared to healthy controls, indicating that full biomechanical restoration may not occur in all individuals [64], with some studies also reporting alterations in distal gait parameters such as foot progression angle and center of pressure trajectory during running [66]. The current findings further support the notion that peak knee flexion angle recovery follows a dynamic trajectory characterized by early

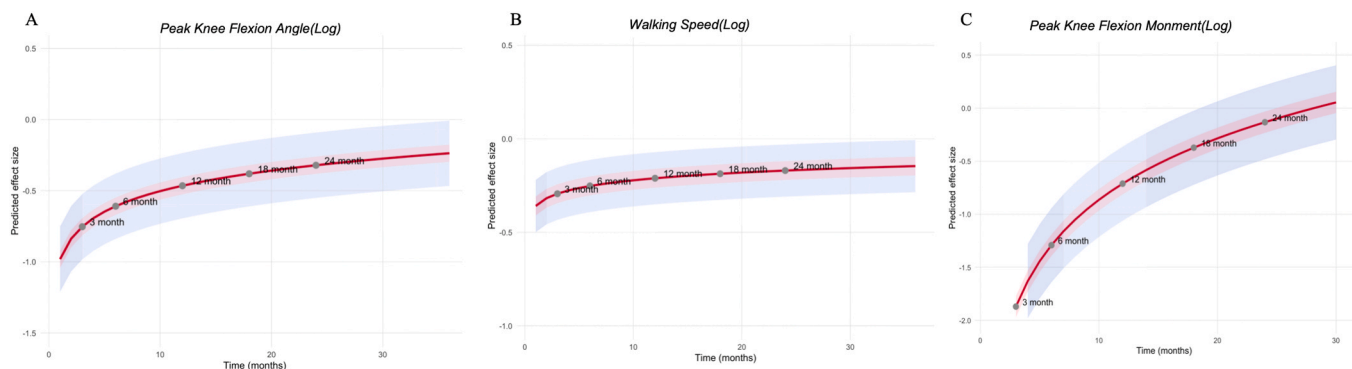


Fig. 3. Trajectory of Gait Recovery at Key Postoperative Time Points. Predicted recovery trends for gait parameters across multiple postoperative timepoints. The figure illustrates the point at which group differences become statistically non-significant, indicating functional normalization.

limitations followed by mid-term normalization, although long-term deviations may persist in some cases.

4.2.2. Temporal changes in knee flexion moment

A similar temporal pattern was observed for peak knee flexion moment. At 3 months postoperatively, ACLR patients exhibited significantly reduced peak knee flexion moments compared to healthy controls, reflecting early postoperative deficits in neuromuscular output and potential quadriceps avoidance behaviors [50]. The differences between groups gradually decreased over time, becoming statistically non-significant by approximately 10.1 months, suggesting mid-term recovery. This trend aligns with previous studies, which have reported an initial postoperative decline in peak knee joint moments followed by gradual recovery toward normal levels [11]. Our review further reinforces the characterization of peak knee flexion moment recovery as exhibiting early postoperative limitations followed by mid-term recovery.

4.2.3. Temporal changes in walking speed

In contrast to the aforementioned parameters, walking speed did not significantly differ between ACLR patients and healthy controls at any critical postoperative time points, indicating a relatively rapid recovery of walking speed postoperatively. This finding concurs with previous research, which reported no significant differences in walking speeds between ACLR patients and controls within 4–15 months post-surgery [59]. It is noteworthy, however, that subtle gait adjustments, potentially stemming from postoperative neuromuscular control changes [51], such as asymmetries or alterations in specific gait phases [52], may still occur despite the lack of overt differences in walking speed. Consequently, when assessing the recovery of spatiotemporal gait parameters, attention should focus on gait symmetry and critical gait phases in ACLR patients to more comprehensively reflect their functional recovery status.

Although walking speed did not differ significantly between ACLR patients and healthy controls in our analysis, this may not necessarily indicate full walking speed recovery. Previous studies have suggested that patients may adopt compensatory strategies—such as reducing stride length or increasing cadence—to maintain a “normal” walking speed despite persistent neuromuscular deficits and inter-limb asymmetries [7,60]. These adjustments may mask underlying functional limitations. Thus, walking speed alone may not be sensitive enough to detect subtle gait asymmetries or functional limitations, and should be interpreted in conjunction with other gait parameters.

4.3. Limitations and future directions

Given the limited and inconsistent reporting of acute postoperative gait data, this study included only ACLR patients assessed at ≥ 3 months post-surgery, potentially missing early-phase gait changes. Moreover, uneven follow-up time distributions and small sample sizes at certain intervals may have affected the precision of effect estimates.

Despite applying a multilevel modeling framework with continuous autoregressive covariance and robust variance estimation to account for within-study correlations, notable heterogeneity remained—particularly for peak knee flexion moment ($\tau^2 = 1.15$). This variability may reflect differences in graft types, rehabilitation protocols, or follow-up durations. However, insufficient reporting of these variables precluded stratified analyses, limiting our ability to identify sources of heterogeneity. Additionally, the lack of standardized reporting for certain gait parameters (e.g., phase-specific metrics) limited more comprehensive exploration of motor control mechanisms.

Although including studies that used contralateral limbs as controls may introduce additional sources of heterogeneity, it is also important to acknowledge that excluding them and limiting the analysis to between-group comparisons does not eliminate heterogeneity either.

Future research should focus on conducting high-quality,

prospective studies with extended follow-up periods, covering gait changes across all postoperative phases following ACL reconstruction. While the current study focused exclusively on gait-related parameters, future work should aim to provide a more comprehensive evaluation of knee function. Emphasis should be placed on detailed functional metrics, such as anterior-posterior (AP) joint stability, muscle strength, and limb symmetry indices, to enable a more holistic assessment of functional recovery.

5. Conclusion

This longitudinal multilevel meta-analysis synthesized evidence across multiple postoperative time points to delineate the recovery trajectories of key gait parameters following ACL reconstruction. At ≥ 3 months post-ACLR, patients demonstrated significantly reduced peak knee flexion angle and peak knee flexion moment compared to healthy controls. These deficits progressively resolved, reaching statistical normalization at approximately 16.2 and 10.1 months, respectively. Walking speed did not significantly differ at any postoperative time point but may reflect compensatory strategies employed to maintain normal gait. These findings highlight the importance of targeting quadriceps strengthening and knee flexion restoration during early-phase rehabilitation, particularly within the first 10–16 months, to address persistent biomechanical deficits.

Author contributions

Shiwei Chen, Han Gong: conducted the experimental design and execution, carried out data analysis, crafted figures and tables, wrote and reviewed the initial article drafts, and gave approval for the final version.

Chennan Lyu, Kehan Li: Carried out the experiments, designed figures and tables, and gave approval to the final draft.

Shazlin Shaharudin: designed the experiments, authored and reviewed the initial article drafts, and gave approval for the final version.

CRediT authorship contribution statement

Chennan Lyu: Methodology, Data curation, Conceptualization. **Kehan Li:** Software, Resources, Methodology, Data curation. **Shazlin Shaharudin:** Writing – review & editing, Supervision, Methodology, Formal analysis. **Shiwei Chen:** Writing – review & editing, Writing – original draft, Methodology, Investigation, Formal analysis, Data curation. **Han Gong:** Writing – original draft, Methodology, Investigation, Formal analysis, Conceptualization.

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Previous presentation

This research has not been previously presented in any form, including conferences, symposiums, or online preprint servers.

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Declaration of Competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence

the work reported in this paper.

Appendix A. Supporting information

Supplementary data associated with this article can be found in the online version at [doi:10.1016/j.gaitpost.2025.109972](https://doi.org/10.1016/j.gaitpost.2025.109972).

Data availability

Code availability: Full dataset and analysis scripts available from <https://github.com/Shiwei-Lab/ACLR-Gait-Meta2025/tree/main/scripts>

Availability of data and material: Full dataset and analysis scripts available from <https://github.com/Shiwei-Lab/ACLR-Gait-Meta2025/tree/main/data>

Supplementary File: The online version contains supplementary material available at <https://github.com/Shiwei-Lab/ACLR-Gait-Meta2025/tree/main/supplement>

Modeling Results: Detailed results from model testing, selection and final estimation processes are available at:

<https://github.com/Shiwei-Lab/ACLR-Gait-Meta2025/tree/main/Results%20Output>

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